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Betsi Cadwaladr
University Health Board

ANAESTHESIA, CRITICAL CARE & PAIN MANAGEMENT DIRECTORATE

YSBTY GWYNEDD

MANAGEMENT OF PATIENTS WITH ISOLATED CHEST TRAUMA

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Author(s):	Dr Stephan Clements	**Author(s) title:**	Consultant Anaesthetist
Responsible dept / director:	Department of Anaesthesia, BCUHB (West) Dr Ed Farley-Hills Clinical Lead of Anaesthesia, BCUHB (West)		
Approved by:	Department of Anaesthesia, STAG, Acute Pain Team, BCUHB (West),		
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Review	**Purpose of Issue/Description of current changes:** Ensure best practice for managing patients with rib fractures and other isolated chest trauma presenting to Betsi Cadwaladr University Health Board (West)

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1. Scope

This guideline is intended as a reference for all medical staff seeking advice regarding management of patients with rib fractures.

2. Key Points

- Multiple rib fractures are often associated with underlying pulmonary contusion.
- Early CT scanning with contrast should be immediately available to define chest wall injuries. 3D surface rendered images should be available within 24 hours, if rib fractures are identified.
- Patients with multiple fractured ribs are at risk of respiratory compromise, if they are unable to cough and clear secretions due to pain. Respiratory compromise may not be seen for 2-3 days after injury and should be anticipated based on risk scoring.
- The risk of harm increases with increasing numbers of fractured ribs with 3 or more considered as a high risk injury.
- The impact of associated fractures to the clavicle, scapula, sternum and thoracic spine should not be underestimated as these can produce a 'functional flail' in association with a relatively small number of rib fractures, causing a greater degree of morbidity than might otherwise be anticipated.
- Early effective pain management is necessary to reduce the risk of respiratory complications and facilitate rehabilitation and recovery. Involve the acute pain team and on call Anaesthetist early.

3. Good Practice Principles

- Risk Stratification
- Early and effective pain management (anaesthetic & acute pain team involvement)
- Early referral to physiotherapy
- Close monitoring for signs of deterioration
- Referral for rib fracture fixation (where appropriate)



3.1 Risk Stratification

Complete the STUMBL risk score tool (Appendix 1) for all patients presenting with rib fractures. It allows patients to be stratified into three groups, supporting the ED clinician in deciding whether to discharge, admit to a ward or refer to critical care.

Risk score	Probability of complications
0-10	13%
11-15	29%
16-20	52%
21-25	70%
26-30	80%
31+	88%

Total Score 0-11:	Consider discharge home with patient advice leaflet and analgesia Recommended GP follow up in 4 weeks.
Total Score 12-26:	Consider admission to a ward for Observation, Analgesia & Physiotherapy
Total Score ≥27:	Consider Critical Care Admission

3.2 Care Bundle Implementation

3.21 Within 1 hour of Diagnosis

Within 1 hour of diagnosis aim to
Start incentive spirometry
Start continuous SpO2 monitoring
Provide supplemental O2, if required
Pain score using verbal rating score

3.22 Early & Effective Pain Management

Effective analgesia starts in the Emergency Department. Use the following analgesia guide.

Mild Pain

Regular Paracetamol 1g 4-6 hourly IV/PO (750mg if <50 kg) Initial dose should preferably be IV loading dose
Regular Ibuprofen 600mg 8 hourly PO Caution: Asthma, gastric irritation, elderly & renal disease
± Codeine 30-60mg PO 4-6 hourly
OR Tramadol 50-100mg PO/IV 4-6 hourly
(avoid if risk of Delirium eg. elderly)
± Oral Morphine (adjust by age*) 2 hourly PRN for breakthrough pain

Moderate/Severe Pain

1. Initial: IV Morphine 0.1-0.2mg/kg titrated to effect
2. Regular Paracetamol & Ibuprofen± PPI. Caution: Asthma, gastric irritation, elderly & renal disease
3. Either:
   a. Serratous Anterior / Erector Spinae Plane Block *(single shot)* or
   b. Apply Lidocaine 5% patch over painful area. *(remove every 12 hours & replace 12 hours later)*
4. Opioids (adjust by age*):
   a. If eGFR >30ml/min: MST 10-20mg bd + Oral morphine PRN
   b. If eGFR <30ml/min: Longtec 5-10mg bd + Shortec 2 hourly PRN

OR

Opioid Patient Controlled Analgesia



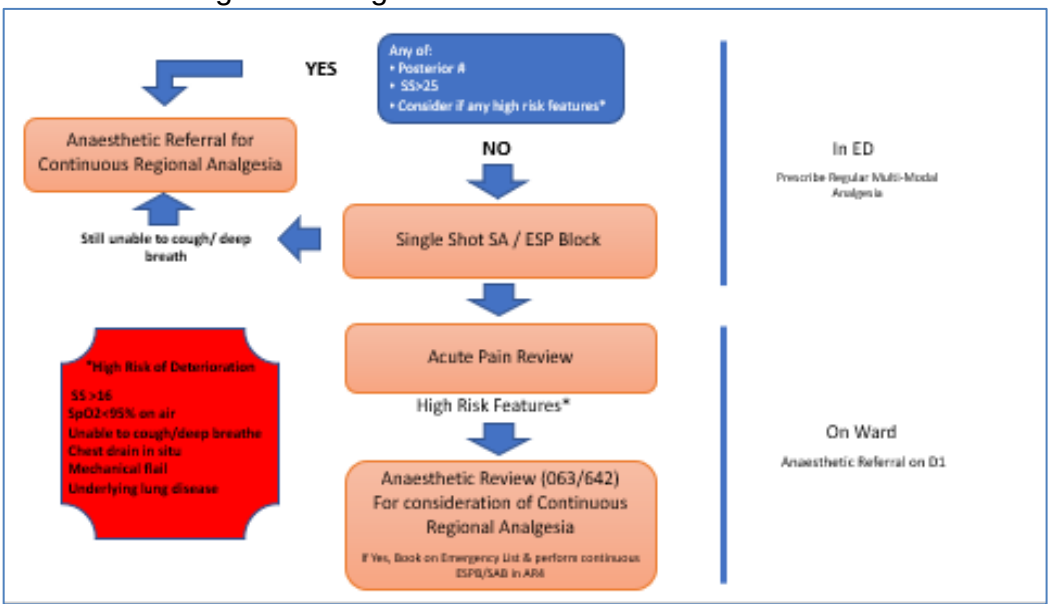
5. Regional Anaesthesia via catheter insertion and elastomeric pump +/- bolus (see guideline YGCG1105)
6. Consider Surgical Fixation

N.B. Whenever opioids are prescribed, also prescribe Antiemetics PRN (+ regular, if PCA), Naloxone PRN and consider regular laxatives.

*Age adjusted oral opioid:			
Oral Morphine PRN 2 hrly:		Oral Oxycodone IR PRN 2 hrly:	
Age (years)	Dose (mg)	Age (years)	Dose (mg)
18-59	20-30mg	18-59	10-15mg
60-69	10-20mg	60-69	5-10mg
70-89	5-10mg	70-89	2.5-5mg
>89	2.5-5mg	>89	1-2.5mg

3.23 Within 24 hours of Diagnosis

- Acute pain team review- refer all in-patients via Bleep 115.
- Physiotherapy review (see 3.24)
- Initiate continuous regional analgesia:



- Consider referring directly to Anaesthetics (Bleep 063) for consideration of Regional Anaesthesia, if any of the following high risk features:
  - Stumble Risk score ≥16
  - Unable to breathe deeply or cough without moderate or severe pain (Dynamic Pain Scores 2-3).
  - Mechanical flail segment
  - Chest drain in situ
  - Underlying lung disease
  - SpO2<95% on air

*Regional technique options include:

- Epidural (Bilateral fractures),
- Continuous Paravertebral block (consider if chest drain in situ or scapula fracture)
- Continuous Erector Spinae plane / MPT block (suitable for all rib fractures)
- PEC2 or Serratous Anterior plane block (suitable for anterolateral rib fractures).
- Intercostal Nerve Blocks (For rescue analgesia only)

3.24 Physiotherapy

Respiratory Physiotherapy and mobilisation should be commenced as soon as possible to prevent deterioration including out of hours, if necessary.

Refer all in-patients for physiotherapy.

Principles:

- Regular deep breathing exercises (consider incentive spirometer) followed by coughing.
- Promote early mobilisation provided no management restrictions.
- All oxygen should be humidified
- Patient education. Explain why the exercises are important & provide them with written information (*Appendix 1- please print out from departmental website*)

3.25 Close Monitoring For Signs Of Deterioration

Respiratory compromise may not be seen for 2-3 days after injury and should be anticipated based on risk scoring the patient location be considered accordingly.
Have a low threshold for involving the on call Anaesthetist, critical care, physiotherapy and acute pain team.

3.3 Referral For Rib Fixation

- **Suggested Indications:**

ACUTE-
- Over 40 year of age (Not an absolute cut-off)
- 2 or more Type B/C rib fractures
- Chest wall deformity
- Flail segments
- Poor pain control

Timing of surgery - Preferably within first 72hrs

CHRONIC-
- Significant displacement where non-or malunion likely to be problematic
- Chronic pain

Timing of surgery - Anytime

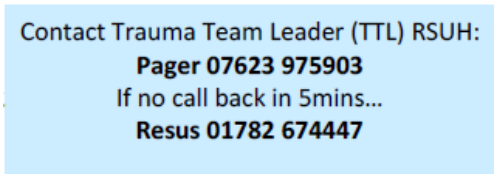
N/B. Type B/C rib fractures: 2mm or more displacement (on CT).

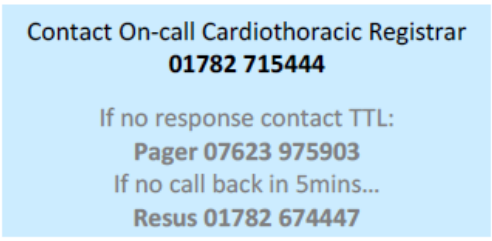
- **Patients likely to be Unsuitable for Fixation:**
  - 1st, 2nd, 3rd, 11th and 12th rib fractures
  - Undisplaced or minimally displaced fractures with no/minimal pain
  - Associated severe head or spinal injury which is not correctable or has not yet been corrected.
  - Most radiological flails do not need any fixation. Rather, it is the presence of Clinical Flail which warrants referral.

- **Who to Refer to:**
  - Patients with multiple injuries or those patients in ED or Critical Care, refer to Trauma Team leader (TTL) at RSUH.

- For patients with **isolated thoracic trauma referral must be made online via the "Refer-a-patient" portal**, then telephone call to on-call Cardiothoracic registrar to inform them of referral.

https://www.referapatient.org/





4 References

- Stoke-on Trent Thoracic Assessment Referral and Transfer Pathway, September 2020
- https://www.nice.org.uk/guidance/ipg361/resources/insertion-of-metal-rib-reinforcements-tostabilise-a-flail-chest-wall-1899867689121733
- Battle C, Hutchings HA, Driscoll T*, et al.* A multicentre randomised feasibility STUdy evaluating the impact of a prognostic model for Management of BLunt chest wall trauma patients: STUMBL Trial. *BMJ Open* 2019;9**:**e029187.

Appendix 1: STUMBL Risk Score

STUMBL
STUdy of the Management of BLunt chest wall trauma

GIG CYMRU NHS WALES
Bwrdd Iechyd Prifysgol Abertawe Bro Morgannwg University Health Board

R&D
RESEARCH & DEVELOPMENT

Swansea Blunt Chest Wall Trauma Score

NB: Circle the score for each risk factor in the table and total all scores

	Patient data	Corresponding risk score
Age	10-19 20-29 30-39 40-49 50-59 60-69 70-79 80-89 90-99 100-109	1 2 3 4 5 6 7 8 9 10
Number of rib fractures	0 1 2 3 4 5 6 7 8 9 10	0 3 6 9 12 15 18 21 24 27 30
Pre-injury anticoagulants	No Yes	0 4
Chronic lung disease	No Yes	0 5
Oxygen saturation levels	100-95% 90-94% 85-89% 80-84% 75-79% 70-74%	0 2 4 6 8 10

Attach Patient Hospital Sticker Here

Total Score

_____

Risk score	Probability of complications
0-10	13%
11-15	29%
16-20	52%
21-25	70%
26-30	80%
31+	88%

Total Score 0-11: Consider discharge home with advice leaflet and analgesia

Total Score 12-26: Consider admission to a ward for observation, analgesia and physiotherapy

Total Score ≥27: Consider ICU management

Actual management decision: (*PLEASE CIRCLE*): **HOME** **WARD** **HDU / ICU**

Physiotherapy: Broken Ribs

You have been given this information because you have been diagnosed with broken ribs. Your rib injuries should heal themselves, though this may take as long as two or three months. You may experience significant pain and this may increase for the first few days after your injury.

The importance of deep breathing and coughing exercises with broken ribs:

Most problems after breaking your ribs will relate to your lungs. Due to pain around your chest, your breathing may be shallow, and your cough may be weak. This can cause phlegm to build up in the lungs, which can lead to a chest infection or pneumonia. You will be advised to do deep breathing exercises (see below) by your medical team, nursing team or physiotherapist. Please do these exercises regularly to minimise the risk of you getting a chest infection.

Breathing exercises

- Carry out your breathing exercises each waking hour throughout your hospital stay.
- Your nurse or physiotherapist will show you the positions in which to do the exercises. This will usually be sitting, in the upright position if your other injuries allow this.
- Breathe slowly in through your nose and out through your mouth. Breathe in fully, drawing air to the bottom of your lungs. **Do this six times, holding every third breath for five seconds.**
- Cough deeply from your stomach to your throat.
- You may or may not cough up some phlegm.
- If you cough up some phlegm spit it into a pot or tissue and repeat the cycle until you are no longer coughing up any phlegm.
- **Repeat the exercises every waking hour.**
- If pain is preventing you from performing these exercises regularly, do inform your nurse so that you receive the right amount and type of pain relief.

Walking (under guidance from your medical team/ nursing team/ physiotherapist)

When your condition allows, nursing staff or the physiotherapist will help you to sit in a chair and to walk. This may be on the day of your hospital admission. You may have oxygen, drips and drains attached but this should not stop you getting out of bed with support from staff.
Once your ability to sit out in a chair and walk on the ward by yourself has been assessed, aim to sit out at regular intervals and complete regular short walks throughout the day. Gradually increase the distance as you are able. If you have any concerns or queries regarding your exercises then please ask your nurse/ physiotherapist for advice.

Ffisiotherapi: Torasgwrn Asennau

Rydych wedi derbyn yr wybodaeth hon oherwydd eich bod wedi derbyn diagnosis o dorasgwrn asennau. Dylai'ch anaf i'r asennau wella ei hunain, ond gall hyn gymryd hyd at 2-3 mis. Mae'n bosibl byddwch yn cael dipyn o boen, a gall hyn gynyddu am yr ychydig ddyddiau cyntaf ar ôl eich anaf.

Pwysigrwydd ymarferion anadlu dwfn a pheswch gyda thorasgwrn asennau:

Bydd y rhan fwyaf o broblemau yn dilyn torasgwrn asennau yn ymwneud â'ch ysgyfaint. Oherwydd y boen o gwmpas eich brest, mae'n bosibl bydd eich anadlu'n ysgafn, a bydd eich peswch yn wan. Gall hyn achosi i fflem adeiladu yn yr ysgyfaint, a all arwain at haint y frest neu niwmonia. Fe'ch cynghorir i ymgymryd ag ymarferion anadlu'n ddwfn (gweler isod) gan eich tîm meddygol, tîm nyrsio neu ffisiotherapydd. Dylech wneud yr ymarferion hyn yn rheolaidd i leihau'r risg o haint ar y frest.

Ymarferion anadlu

- Dylech wneud eich ymarferion anadlu bob awr y byddwch yn effro drwy gydol eich arhosiad ysbyty.
- Bydd eich nyrs neu ffisiotherapydd yn dangos y gwahanol osgo dylech eu defnyddio i wneud yr ymarferion. Fel arfer bydd y rhain yn cynnwys eistedd, ar osgo syth, os bydd eich anafiadau eraill yn eich caniatáu i wneud hyn.
- Anadlwch yn araf drwy eich trwyn ac allan drwy eich ceg. Anadlwch yn llawn, gan dynnu'r anadl i waelod eich ysgyfaint. **Gwnewch hyn chwe gwaith, gan ddal bob trydydd anadl am bum eiliad.**
- Dylech beswch yn ddwfn o'ch stumog i'ch gwddf.
- Mae'n bosibl byddwch yn peswch ychydig o fflem, neu efallai ddim.
- Os byddwch yn peswch fflem, dylech ei boeri i botyn neu hances bapur, ailadroddwch y cylch hyd nes na fyddwch yn peswch unrhyw fflem.
- **Ailadroddwch yr ymarferiad bob awr byddwch yn effro.**
- Os yw'r boen yn eich rhwystro rhag perfformio'r ymarferion rhain yn rheolaidd, rhowch wybod i'ch nyrs fel eich bod yn derbyn y dos a'r math o boen laddwr cywir.

Cerdded (gan ddefnyddio canllawiau eich tîm meddygol/tîm nyrsio/ffisiotherapydd)

Pan fydd eich cyflwr yn eich galluogi, bydd y staff nyrsio neu'r ffisiotherapydd yn eich helpu i eistedd ar gadair a cherdded. Efallai bydd hyn yn digwydd ar ddiwrnod eich derbyn i'r ysbyty. Efallai byddwch yn cael ocsigen a bydd dripiau a draeniau wedi'u cysylltu arnoch, ond ni ddylai hyn eich rhwystro rhag codi o'r gwely gyda chefnogaeth gan staff.
Pan fydd eich gallu i eistedd ar gadair a cherdded ar y ward heb gymorth wedi'u hasesu, anelwch i eistedd allan a mynd am dro bach yn rheolaidd drwy gydol y dydd. Yn raddol, cynyddwch y pellter. Os bydd gennych unrhyw bryderon neu ymholiadau ynghylch eich ymarferion, gofynnwch i'ch nyrs/ffisiotherapydd am gyngor.

Appendix 3: Referral Check List

Referral Checklist	Comments:		
Referring Consultant and contact number			
Patient location and ward contact number			
Name / DOB / NHS no			
Date / Time / mechanism of injury			
STUMBL Score (see appendix1)			
Other injuries or fractures (e.g. scapula, clavicle, sternum or spine) and other systems	**Isolated chest trauma excludes thoracic vertebral trauma which should be referred via severe polytrauma pathway**		
Age If > 65 calculate Clinical Frailty Score (see appendix 3)	**Very elderly ≥80 & high frailty score ≥ 7 discuss with MTC**		
GCS >13, ≤ 13	**(If ≤ 13 , Refer to MTC-TTL not CTS)**		
Type of Flail Chest Establish from 3D Recon of CT thorax	**A / B / C / D** *(UHNM Classification see Appendix 2)*		
Number of rib fractures left /right Specify presence, side and size of pleural complications	**No of ribs #**	**Right**	**Left**
	Haemothorax		
	Pneumothorax		
	Tension Pneumothorax		
	Chest drain inserted		
PaO2 and FiO2? Intubated? CPAP?			
NEWS2 score GCS	**At presentation**	**Current after pain relief**	
Co-morbid disease e.g. COPD/ILD/CF/BMI >30 /OSA Cardiac /Renal/ Neurological conditions / liver injury/ dysfunction / Pre-injury anticoagulation			
Covid-19 swab			

START Pathway Sept 2020